

Solving rational inequalities algebraically

Key Points

$$f(x) = \frac{N(x)}{D(x)}$$

Critical points:

To solve $f(x) = 0$ we need only solve $N(x) = 0$. The undefined points of $f(x)$ are given by $D(x) = 0$

To solve the inequality we need to multiply by $D(x)$ and consider a case by case basis of $D(x) > 0$ and $D(x) < 0$. Or multiply a square.

Alternatively a table can be used by also considering $N(x) > 0$ and $N(x) < 0$

Basic Skills

Q1. For the rational function $f(x) = \frac{2x+5}{6-x}$

- a) Write down $N(x)$ and $D(x)$
- b) State the critical points i.e. when $N(x) = 0$ or $D(x) = 0$
- c) Complete the following table

Positive/Negative	$x < -2.5$	$-2.5 < x < 6$	$x > 6$
$2x + 5$	—	+	
$6 - x$			
$f(x)$			

Q2. Make a positive/negative table for $f(x) = \frac{6x-36}{2x-4}$

Competency Skills

Q3. a) Make a positive/negative table for $f(x) = \frac{x^2-9}{10-2x}$

b) Hence solve the inequality $\frac{x^2-9}{10-2x} \geq 0$

c) Hence solve the inequality $\frac{x^2-9}{10-2x} < 0$

Q4. a) Solve the inequality $\frac{x+3}{(x-2)(5-x)} \leq 0$

b) Hence solve $\frac{5}{3(x-2)} \leq \frac{8}{3(x-5)}$

Q5. Solve the inequality $x + 5 \geq \frac{1}{x}$

Q6. Solve the inequality $\frac{x+1}{x+2} > 2x + 1$

Advanced Skills

Q7. a) Draw the graphs $y = \frac{1}{x-2}$ and $y = 5x + 2$ stating the points of intersection

b) Hence, or otherwise, solve $5x + 2 \geq \frac{1}{x-2}$

Q8. Solve the inequality $\frac{x^2-2x+1}{x^2-6x+8} \geq 0$

Q9. Explain why solving an inequality is hard but solving equality is easy

Q10. Solve the inequality $\frac{5x+1}{2x} < \frac{8}{x+1}$

Q11. Explain why solving the inequality $\frac{x^2-6x-5}{x^2+5} \geq 2$ is relatively easy compared to others

Extension

Q12. Come up with another possible method to solve rational inequalities rather than dealing with a case by case basis, multiplying by a square or using a table

Hint: Look at Q7

For more help, please visit www.exampaperspractice.co.uk